

An Optimized AI-Driven Framework for Early and Reliable Dental Caries Detection Using ResNet-50

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Abstract—Dental caries detection from panoramic radiographs is a complex diagnostic task requiring precise interpretation of subtle structural variations. This study proposes an optimized binary classification framework using ResNet-50 for automated caries detection. The dataset, originally in YOLO detection format, was restructured into a balanced classification dataset to reduce class bias. Unlike conventional approaches that freeze most pre-trained layers, the proposed method selectively unfreezes deeper convolutional blocks to improve domain-specific feature learning. Epoch tuning was also applied to ensure stable convergence and reduce overfitting. The model achieved a classification accuracy of 93%, outperforming baseline implementations. Grad-CAM visualization further enhances interpretability by highlighting diagnostically relevant regions. These results demonstrate that effective fine-tuning and balanced data preparation significantly improve the reliability of deep learning-based caries detection systems.

Keywords— Dental caries, ResNet-50, transfer learning, panoramic radiography, Grad-CAM.

1. INTRODUCTION

Dental caries is one of the most prevalent oral diseases worldwide and requires early detection to prevent structural tooth damage and reduce invasive treatments. Panoramic radiography is widely used in dental diagnostics due to its ability to provide a comprehensive view of the maxillofacial region. However, manual interpretation of radiographs is time-consuming and often subject to variability based on clinician experience and image quality. These challenges highlight the need for automated diagnostic systems that can improve consistency and efficiency. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs) with transfer learning, have shown strong performance in medical

image analysis. ResNet-50, with its residual learning framework, enables effective training of deep networks and has been widely applied in radiographic classification tasks. Prior work [1] utilized ResNet-50 combined with Grad-CAM for explainable dental caries detection, achieving an accuracy of 92% while emphasizing model interpretability.

Despite these advancements, model performance remains sensitive to factors such as layer freezing strategies and dataset imbalance, which can limit domain-specific learning and introduce bias. To address these limitations, this study proposes a fine-tuned ResNet-50 framework with selective unfreezing of deeper layers, improved dataset balancing, and optimized training. The proposed approach achieves an accuracy of 93% and employs Grad-CAM to ensure clinically relevant feature interpretation, enhancing the reliability of automated dental caries detection systems.

II. RELATED WORK

Automated dental pathology detection using deep learning has gained significant attention due to the ability of Convolutional Neural Networks (CNNs) to extract hierarchical features from image data. Early studies focused on localized regions such as bitewing radiographs, where Lee et al. demonstrated effective caries classification using CNN-based models [2]. In medical imaging, transfer learning has been widely adopted to address limited dataset sizes by adapting pre-trained models, such as those trained on ImageNet, for domain-specific tasks [3], [4]. Additionally, residual networks, such as ResNet-50, enable deeper feature learning by mitigating vanishing gradient issues through skip connections [5]. In dental caries detection, Oztekin et al. proposed a ResNet-50-based

model with Grad-CAM visualization, achieving 92% accuracy while providing interpretability of predictions [1]. However, their approach utilized a fixed transfer learning configuration without exploring deeper fine-tuning strategies. Other studies have highlighted the impact of dataset imbalance on model performance, emphasizing the need for rebalancing techniques to improve sensitivity toward pathological cases [3], [4].

Despite these advancements, the combined effect of fine-tuning depth, dataset balancing, and training stability remains underexplored in dental caries classification. This study addresses these gaps by optimizing fine-tuning strategies, improving dataset balance, and refining training parameters to enhance overall model performance.

III. MATERIALS AND DATASET PREPARATION

A. Dataset Description

The dataset consists of panoramic dental radiographs originally annotated in YOLO object detection format, with bounding boxes marking multiple dental conditions. As shown in Fig. 1, the images exhibit variations in quality, anatomical structure, and pathological features, highlighting challenges in consistent feature extraction for automated caries detection.

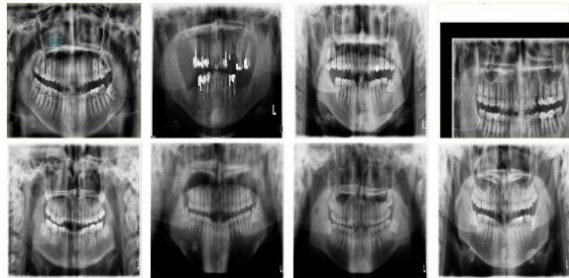


Fig. 1. Representative panoramic radiographs from the dataset used for binary caries classification.

The dataset was reformulated into a binary classification framework consisting of *Caries* and *Non-Caries*. Images containing at least one carious annotation were labeled as caries, while others were classified as non-carious, enabling direct image-level classification. To convert detection-based annotations into classification inputs, bounding box information was used to extract tooth-level regions. As shown in Fig. 2, these cropped regions focus on diagnostically

relevant structures, reducing background noise and improving feature localization. The extracted samples were subsequently resized and normalized prior to training.

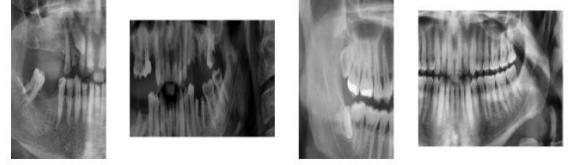


Fig. 2. Example cropped tooth-level samples extracted from YOLO-annotated panoramic radiographs for binary classification.

B. Dataset Distribution and Balancing Strategy

Medical imaging datasets frequently exhibit class imbalance, where pathological samples are underrepresented relative to normal cases. To prevent bias toward the dominant class, a structured dataset balancing strategy was implemented. Data augmentation techniques were applied selectively within the training set to equalize the number of caries and non-carious samples while preserving clinically relevant features.

Phase	Original Caries	Original Non-Caries	Augmented Caries	Augmented Non-Caries
Train	860	740	6635	6635
Test	300	300	300	300
Total	2200	1600	13870	6935

Table I: Distribution of original and augmented samples across training and testing sets.

As shown in Table I, the training dataset was balanced through augmentation to ensure equal representation of both classes. The testing dataset remained unaltered to provide an unbiased evaluation of model performance. This strategy reduces prediction bias and improves the robustness of classification metrics.

IV. PROPOSED METHODOLOGY

A. ResNet-50 Architecture

ResNet-50 was selected due to its residual learning framework, which enables effective training of deep convolutional networks through identity shortcut connections. The architecture consists of 50 weighted layers organized into convolutional blocks with residual mappings that mitigate vanishing gradient issues. Residual connections allow the network to learn feature refinements rather than complete transformations, improving convergence stability in

deeper models. Pre-trained ImageNet weights were utilized to initialize the model parameters. Leveraging pre-trained representations allows efficient adaptation to domain-specific radiographic features while reducing the computational cost associated with training from scratch.

B. Transfer Learning Configuration

Unlike conventional approaches that freeze a large portion of pre-trained layers, the proposed framework selectively unfreezes deeper convolutional blocks to enhance domain adaptation. Freezing excessive layers may restrict the model's ability to capture radiographic texture variations specific to dental imagery. Therefore, only early feature extraction layers were retained as fixed, while deeper layers were fine-tuned to learn domain-relevant representations. This strategy allows the model to preserve generic low-level features such as edges and gradients while adapting higher-level representations to the specific patterns of carious lesions.

C. Training Optimization

Model training was conducted using an optimized epoch tuning strategy to ensure stable convergence and avoid overfitting or underfitting. A dataset balancing approach was applied to maintain equal representation of caries and non-caries samples, improving sensitivity to pathological cases. Input images were resized to match ResNet-50 requirements and normalized to enhance training stability. Additionally, the final fully connected layer was modified to produce binary classification outputs.

D. Model Interpretability Using Grad-CAM

To enhance clinical transparency, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visualize salient image regions influencing classification decisions. Grad-CAM generates heatmaps by utilizing gradient information flowing into the final convolutional layer, thereby identifying discriminative areas responsible for predictions.

This interpretability mechanism ensures that the model's predictions are aligned with clinically meaningful structures rather than background

artifacts, supporting its applicability in real-world dental diagnostics.

V. EXPERIMENTAL SETUP

A. Implementation Environment

All experiments were conducted using the PyTorch deep learning framework. The implementation was executed in the Google Colaboratory environment. Training was performed using NVIDIA GPU acceleration when available; however, additional validation and inference were also verified under CPU settings to ensure reproducibility across computational setups.

The model utilized pre-trained ImageNet weights as initialization, enabling efficient convergence during transfer learning.

B. Training Configuration

The modified ResNet-50 architecture was trained for 20 epochs using a two-stage strategy. In the feature extraction phase, the final fully connected layer was replaced for binary classification while earlier layers were frozen. In the fine-tuning phase, selected deeper layers were unfrozen to enable domain-specific adaptation for radiographic features. The model was trained using cross-entropy loss and the Adam optimizer with a learning rate of 0.0001, a batch size of 32, and an input resolution of 224×224 pixels. Data normalization followed ImageNet standards, and dataset balancing ensured equal class representation.

C. Evaluation Metrics

Model performance was evaluated using multiple quantitative metrics to ensure a comprehensive assessment, including accuracy, precision, recall (sensitivity), F1-score, confusion matrix, and Receiver Operating Characteristic (ROC) curve with Area Under the Curve (AUC). Accuracy is computed as the ratio of correctly classified samples to the total number of samples, as defined in (1).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. While accuracy provides an overall measure of performance, additional metrics such as precision, recall, and F1-score are essential for evaluating classification effectiveness, particularly in medical diagnostic systems where class imbalance and misclassification costs are critical. Figure 3 illustrates the confusion matrix of the proposed ResNet-50 model, providing a detailed breakdown of prediction outcomes across the two classes. The matrix highlights the model's ability to correctly classify both caries and non-caries samples, while also indicating the distribution of false predictions.

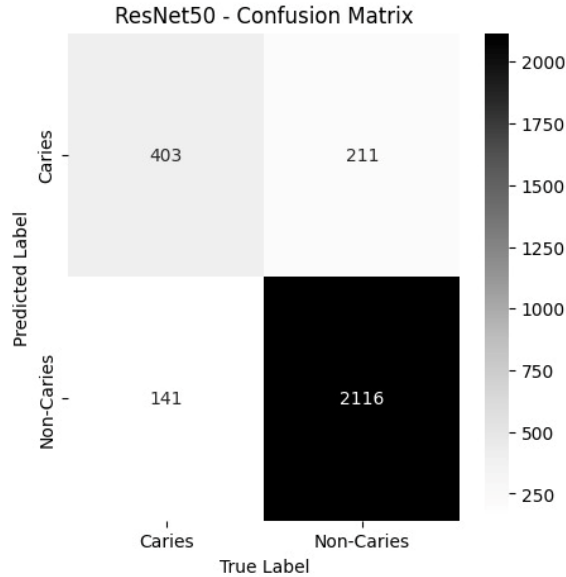


Fig. 3. Confusion matrix of the proposed ResNet-50 model.

VI. RESULTS AND PERFORMANCE ANALYSIS

A. Training Performance

The modified ResNet-50 model demonstrated stable convergence during both the feature extraction and fine-tuning phases. Progressive improvement in training accuracy was observed across epochs, indicating effective learning of discriminative radiographic features. As shown in Fig. 4, the training and validation accuracy exhibit a consistent upward trend across epochs, reflecting stable optimization and minimal overfitting. The two-stage training strategy, involving selective layer unfreezing, contributed to

improved feature specialization for dental imagery. Unlike conventional transfer learning approaches that freeze a majority of layers, controlled fine-tuning of deeper convolutional blocks enhanced domain adaptation without introducing instability.

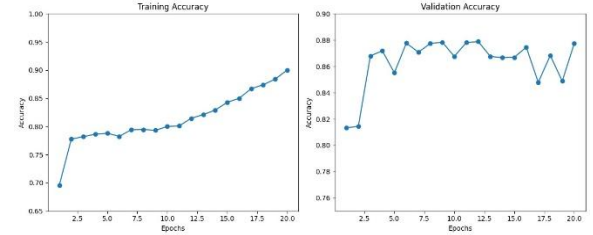


Fig. 4. Training and validation accuracy progression across epochs.

B. Quantitative Performance Metrics

After 20 training epochs, the proposed model achieved an accuracy of 93% on the validation dataset. Performance metrics included a precision of 95.74%, recall of 90.00%, F1-score of 92.78%, and an AUC of 0.94, indicating a strong balance between precision and recall. The confusion matrix further demonstrates robust class-wise discrimination with minimal misclassification, highlighting the reliability of the proposed approach for dental caries detection.

C. Comparative Analysis with Base Model

The baseline study reported an accuracy of approximately 92% using conventional transfer learning, whereas the proposed framework achieved 93%, demonstrating improved classification performance. This gain is attributed to selective layer unfreezing for deeper feature adaptation, improved dataset balancing, optimized epoch tuning, and a structured fine-tuning strategy. Grad-CAM further supports model interpretability and validation. Although the improvement is modest, even small accuracy gains are significant in medical diagnostics, as they enhance the reliability of early detection.

D. Model Interpretability Assessment

Grad-CAM visualization results demonstrate that the model focuses primarily on clinically relevant tooth regions exhibiting structural degradation, rather than surrounding background artifacts. This validates the robustness and reliability of the learned representations.

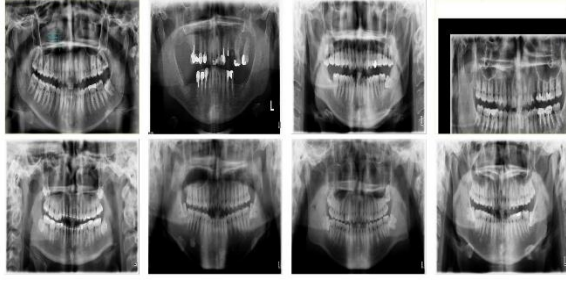


Fig. 5. Grad-CAM visualizations highlighting discriminative regions associated with caries detection.

E. Discussion

The proposed methodology demonstrates that controlled fine-tuning of deep convolutional architectures improves dental radiograph classification without increasing model complexity. The achieved accuracy of 93% highlights the effectiveness of selective transfer learning and balanced dataset preparation. As shown in Table I, the proposed model achieves competitive performance compared to existing approaches, despite using a simpler architecture and a relatively smaller dataset.

Study	Number of Classes	Number of Images	Classifier	Accuracy
Singh and Sehgal [32]	6 (Class I–VI), 3 (Non-caries)	1500	CNN-LSTM	96.00%
Salehi et al. [33]	Enamel, Dentin (4 classes incl. sound)	81	CNN	90.75%
Wang et al. [37]	White-spot lesions, Smashed, Plaque	7200	T-Net CNN	95.45%
Huang and Lee [34]	3 (Non-caries, Enamel, Dentin)	748	ResNet-152	95.21%
Lakshmi and Chitra [35]	2 (Cavity, No cavity)	1900	AlexNet	96.08%
Leo and Reddy [36]	2 (Non-caries, Caries)	480	DNN	96.00%
Zhu et al. [38]	Moderate caries (multi-class)	3127	CariesNet	93.61%
Oztekin et al.	2 (Caries, Non-caries)	13,870	ResNet-50	92.00%
Proposed Method	2 (Caries, Non-caries)	(Your dataset size)	ResNet-50 (Fine-tuned)	93.00%

Table 2. Comparison of our work with some state-of-the-art study techniques (deep learning) developed for automated caries detection

While some methods report higher accuracies, they often rely on larger datasets or more complex models. In contrast, the proposed approach maintains a strong balance between performance and computational efficiency. These findings emphasize that effective fine-tuning and dataset optimization can significantly enhance performance without requiring increased architectural complexity.

VII. LIMITATIONS AND FUTURE WORK

Despite the improved performance, several limitations remain. The dataset, although balanced, is relatively limited, which may affect generalization. Future work should incorporate larger, multi-institutional datasets to improve robustness. The current framework is restricted to binary classification and does not capture different stages of dental caries. Extending the model to multi-class classification would enhance clinical applicability. Additionally, the model was evaluated in a controlled environment; real-world deployment may introduce variability in image quality and patient conditions, requiring further validation and optimization for real-time use.

While Grad-CAM provides interpretability, it remains a post-hoc method and does not ensure complete clinical explainability. Future research may also explore object localization techniques and ensemble learning to improve diagnostic capability and predictive performance.

VIII. CONCLUSION

This study presented an enhanced transfer learning framework for automated dental caries detection using a modified ResNet-50 architecture. Through selective layer unfreezing, improved dataset balancing, and structured fine-tuning, the model achieved an accuracy of 93%. The results demonstrate that controlled deep feature adaptation improves domain-specific learning in dental radiographs. Grad-CAM visualization further validated the model's interpretability by highlighting clinically relevant regions.

Although the improvement over baseline methods is modest, even small gains are significant in clinical

diagnostics. The findings confirm that optimized fine-tuning of pre-trained networks is an effective and efficient approach for medical image classification, contributing to a more reliable system for dental caries detection.

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